

A GCN1-independent activator of the kinase GCN2

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eLife Assessment

This study presents a screen for small-molecule activators of the kinase GCN2 that phosphorylates the eukaryotic translation initiation factor 2 alpha (eIF2α) in response to diverse stress stimuli. Among the compounds identified, one stands out as a potent activator that functions independently of GCN1, which is **important** for probing mechanisms of Integrated Stress Response regulation and may have translational relevance in the context of pathogenic GCN2 mutations. While some reviewers found the biochemical analyses **convincing**, others viewed the cellular evidence as limited, particularly with respect to time points, endogenous readouts, and broader cell-type validation, which prevents a clear assessment of the compound's potential potency in a physiological context.

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Abstract

Mutations of *EIF2AK4*, which encodes the eIF2α kinase GCN2, cause a severe inherited form of pulmonary hypertension called pulmonary veno-occlusive disease (PVOD). Some pathogenic variants of GCN2 are amenable to pharmacological reactivation by low concentrations of ATP-pocket binding inhibitors. Kinase inhibition at modestly elevated concentrations limits the clinical utility of these drugs against PVOD. We therefore performed an *in cellulo* chemical screen for GCN2 activators and identified three structurally distinct compounds with low micromolar stimulatory activities. Unlike previously described GCN2 activators, one of these molecules activated GCN2 independently of GCN1. Modelling supported by structure activity screens suggested it binds within the ATP-pocket of GCN2, but unlike existing ligands does not protrude inward into the allosteric pocket or outward into the solvent. This overcomes a key requirement of other GCN2 activators.

Introduction

Phosphorylation of eukaryotic translation initiation factor 2 alpha (eIF2 α) mediates the response to a variety of stresses by regulating translation, metabolism and cell survival (1, 2). In yeast, GCN2 is the sole eIF2 α kinase and triggers the General Control Response to the accumulation of uncharged tRNA and to ribosome stalling (3, 4). Gene duplication gave rise to a family of eIF2 α kinases in multicellular organisms linking eIF2 α phosphorylation to a variety of other stresses. HRI (EIF2AK1) is activated by haem deficiency, oxidative stress and mitochondrial dysfunction (5–7), PKR (EIF2AK2) responds to cytosolic double stranded RNA during viral infections, while PERK (EIF2AK3) is activated by endoplasmic reticulum (ER) stress (2). Mutations affecting these kinases have been linked to the development of disease and so their manipulation by small molecules is being pursued as a means to develop therapies (2, 8–11).

The eIF2 complex recruits initiator methionyl-tRNA to ribosomes commencing protein synthesis (2). This requires eIF2 to bind GTP, which is subsequently hydrolysed. Bound GTP is replenished by the guanine nucleotide exchange factor (GEF) eIF2B (12), but phosphorylated p-eIF2 α binds avidly to eIF2B inhibiting its GEF activity (13). When levels of eIF2-GTP-tRNA-Met subsequently fall, most translation is disfavoured, but some transcripts are translated more efficiently owing to the presence of upstream open reading frames (uORFs) within their 5' untranslated regions (5'UTRs) (1, 14). ATF4 and CHOP are transcription factors upregulated in this manner during stress to induce genes of the Integrated Stress Response (ISR) (1, 2, 14). Eventually, PPP1R15A (also called GADD34), itself an ISR target, dephosphorylates p-eIF2 α to restore the basal state (15, 16).

GCN2 is comprised of five domains—RWD, pseudokinase, kinase, HisRS-like, and C-terminal—and exists constitutively as a homodimer (17, 18). Its kinase activity is stimulated when the RWD domain interacts with colliding ribosomes via the cofactor GCN1 (19–21) or when the HisRS domain interacts with deacylated tRNAs (22, 23). Direct interaction between GCN2 and the ribosome P-stalk can also mediate GCN2 activation during ribosome stalling (24, 25).

Mutations of *EIF2AK4*, which encodes GCN2, cause the aggressive subtypes of familial pulmonary hypertension, pulmonary veno-occlusive disease (PVOD) and pulmonary capillary haemangiomatosis (PCH) (2, 17). Some pathogenic alleles generate hypomorphic variants of GCN2 that we and others have shown can be activated by small molecule inhibitors that bind to the GCN2 ATP pocket, e.g. neratinib and GCN2iB (9, 11, 26). Type-1 inhibitors bind the ATP pocket of the kinase in its active conformation (DFG-Asp-in, α C-helix-in), while type 2 inhibitors bind the inactive conformation (DFG-out, α C-helix-out) (27). Type 1.5 inhibitors, by contrast, bind an intermediate conformation (DFG-Asp-in, α C-helix-out). Several type 1 and 1.5 kinase inhibitors can activate GCN2 at low concentrations while inhibiting at higher concentrations (9, 11, 26). GCN2iB is a relatively selective type 1.5 inhibitor of GCN2 that activates the kinase at submicromolar concentrations (9, 17). It is hypothesised that drug-induced stabilisation of one GCN2 protomer within a dimer is propagated to the second protomer to cause its activation (11, 28). Several inhibitors of other kinases that are in clinical use or development as anticancer agents activate GCN2 in a GCN1-dependent manner, which may contribute to their therapeutic efficacy (26, 29–31). For this reason, GCN2 activators are being sought as potential anticancer drugs.

The complexity of GCN2 activation, involving direct and indirect detection of amino acid deprivation and ribosome stalling, has hampered efforts to reconstitute GCN2 function *in vitro* (32–34). We therefore performed a cell-based chemical screen for GCN2-dependent activators

of the ISR, followed by *in vitro* validation using orthogonal approaches. In doing so, we identified three novel, structurally distinct GCN2 activators. Notably, one of these, designated compound 20, activates GCN2 in a GCN1-independent manner.

Results

Chemical screen for GCN2 activators

We generated clonal Chinese Hamster Ovary (CHO) cell lines expressing an ATF4::NanoLuc reporter comprising the 5' untranslated region (UTR) of human ATF4 mRNA fused to NanoLuc luciferase (**Figure 1A** [↗](#)) ([17](#) [↗](#)). Activation of the ISR could be detected as bioluminescence in response to ISR activators histidinol, tunicamycin and latrunculin A (**Figure 1B** [↗](#)). A high-throughput screen of the BioAscent library of 123,222 drug-like compounds performed by the ALBORADA Drug Discovery Institute, yielded two separate pools of potential ISR inhibitors and activators. Using this knowledge, we assembled a subset of 6,600 compounds from the BioAscent library enriched for potential ISR activators and screened these in 348-well format at 10 μ M for 16 hours. To increase sensitivity, screens were performed with test compound alone or combined with the half-maximal effective concentration (EC₅₀) of histidinol. This yielded 153 compounds that activated the reporter at least three standard deviations above baseline (**Figure 1C** [↗](#)).

The pipeline used to filter hit compounds is illustrated in **Figure 1D** [↗](#). Counter screens were performed against constitutively expressed Nanoluc, to exclude trivial activators of luminescent proteins, leaving 134 hits. Cytotoxicity assays excluded immediately toxic compounds leaving 130. Hits were then re-screened using a second CHO ATF4::Luc2 reporter cell line, shortlisting 89 lead compounds. GCN2-deficient ATF4::NanoLuc CHO reporter cells were generated by introducing the reporter into *Eif2ak4*^{-/-} CHO cells ([24](#) [↗](#)). Thirty compounds were selected based on GCN2 dependency, where stimulation was lost in GCN2 null cells. Liquid chromatography-mass spectrometry (LC-MS) verified compound purity and quality leaving 16 lead compounds of high confidence from the original library stocks.

Orthogonal secondary screen

The ISR can be activated in isolation or in combination with other stress signalling pathways, for example as part of the unfolded protein response (UPR) to ER stress ([2](#) [↗](#)). To distinguish between inducers of ER stress and selective activators of the ISR, we used dual reporter CHO cells that express a fluorescent ISR reporter (CHOP::GFP) and a reporter of IRE1 activity (XBP1::Turquoise) ([35](#) [↗](#)). Histidinol served as a positive control for ISR activation via GCN2, while tunicamycin was used to induced ER stress and activate the UPR. *Eif2ak4*^{-/-} CHO cells were used to assess GCN2 dependence, while *Ppp1r15a*^{-/-} cells, which are impaired in the dephosphorylation of eIF2 α , were used to amplify the effects of ISR activation. Histidinol selectively activated the CHOP::GFP ISR reporter in wildtype and *Ppp1r15a*^{-/-} cells, but not in GCN2 deleted cells, validating this system (**Figure 2A** [↗](#)). As expected, the ISR was exaggerated in the *Ppp1r15a*^{-/-} cells owing to their defective dephosphorylation of eIF2 α . Tunicamycin activated both the CHOP::GFP and XBP1::Turquoise reporter in all three genotypes.

The 16 lead compounds were next examined both in the presence and absence of a submaximal concentration of histidinol. Using this orthogonal approach, three compounds were found to be GCN2 selective activators of the ISR (**Figure 2B** [↗](#) & **C**). Those which induced ER stress or showed minimal GCN2 dependence are not shown. Reporter activation was measured by fluorescence activated cell sorting (FACS) after 18 hours of treatment with 30 μ M compound 18, 20 μ M compound 20, and 7.5 μ M compound 21. The three GCN2 selective activators included compound 18, *N*-(4-*[(furan-2-yl)methyl]sulfamoyl*phenyl)-3-methyl-1-phenyl-1*H*-thieno[2,3-*c*]pyrazole-5-carboxamide,

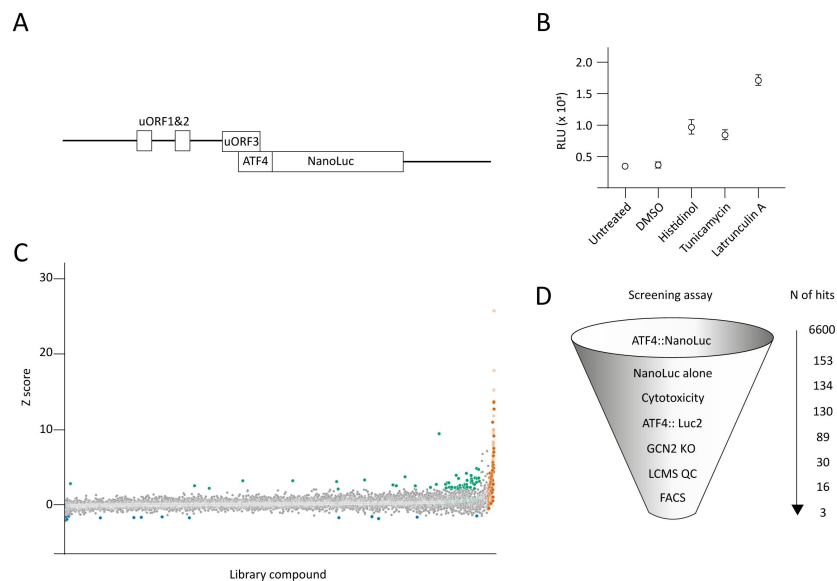


Figure 1.

Chemical Screen of novel GCN2-specific ISR-activating molecules.

(A) Schematic of ATF4::NanoLuc reporter, exploiting the upstream open reading frames (uORFs). Short uORFs 1, 2 and 3 precede the ATF4 translation start site, with uORF3 overlapping the ATF4 coding sequence, but out-of-frame. During the ISR, skipping of uORF3 leads to increased translation initiation at the ATF4 initiator AUG. (B) Relative Light Unit (RLU) of ATF4 signal in CHO stable cell-line overexpressing ATF4::NanoLuc treated with 1mM histidinol, 2.5 μ g/mL tunicamycin, or 1 μ M latrunculin A ($n=3$; $\pm$ SEM). (C) Chemical screen of 6,600 small molecules tested with CHO stable cell-line overexpressing ATF4::NanoLuc. The molecules were treated alone or with submaximal ISR activation with 0.3mM histidinol. All molecules were displayed along the x-axis based on their z-score of relative reporter signal when treated with compound alone (pale symbols). Z-score for treatment with compound and histidinol is also plotted at the same x-axis position (dark symbols). Molecules with z-score > 3 (i.e., over 3 standard deviations from the mean reporter signal) were identified as hits: ISR activator in orange and ISR augments (exaggerating ISR activation by histidinol) in green. (D) Pipeline of molecule short-listing, primary screening using CHO ATF4::NanoLuc reporter (153 hits), NanoLuc alone control (134 hits), selection based on cytotoxicity (130 hits), orthogonal CHO ATF4::FLuc reporter (89 hits), and subsequent GCN2 specificity in GCN2 null reporter cells (30 hits) and compound quality control via LC-MS (16 hits), finally validation by GCN2-specific FACS to a short list of 3 compounds.

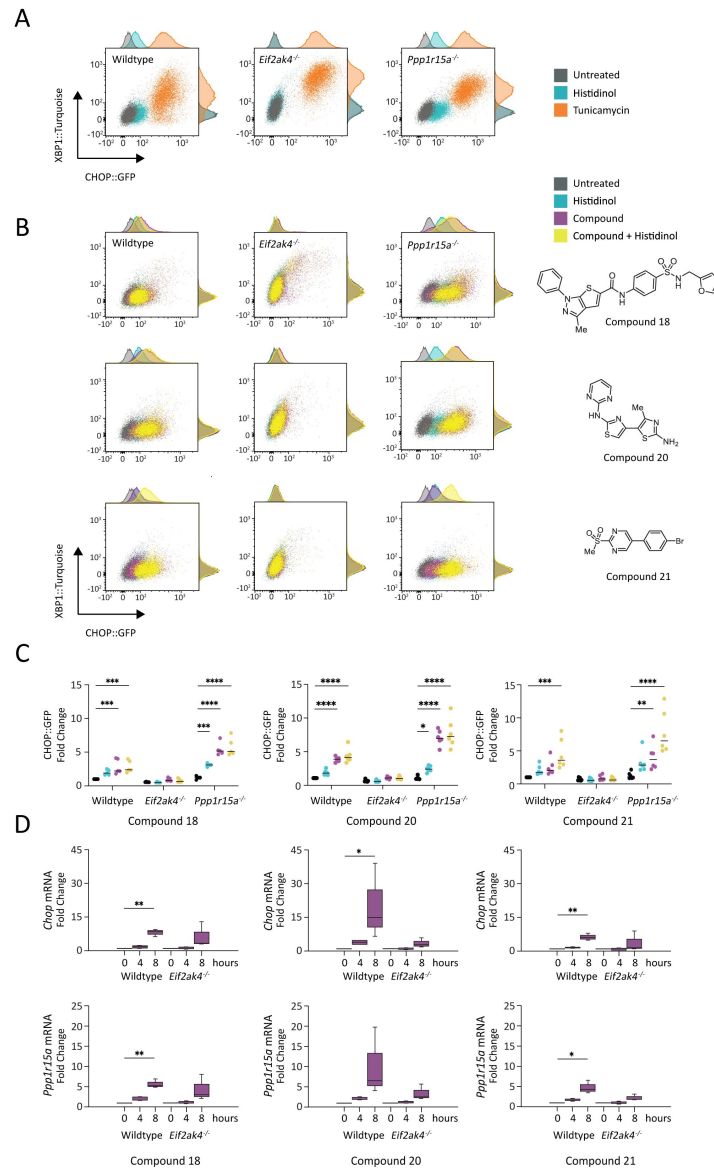


Figure 2.

Shortlist and characterisation of three top hits (compound 18, 20, and 21)

(A) Representative dot plot of flow cytometry data in CHO CHOP::GFP and XBP1::Turquoise dual-reporter wildtype, *Eif2ak4* Δ , and *Ppp1r15a* Δ cells under 18-hour treatment of 0.75mM histidinol (blue), 2 μ g/mL tunicamycin (orange), and untreated control (gray) (n=2). Bi-exponential display was used for the plots. The x-axis represents fluorescence signals at 530nm for GFP (CHOP reporter) and y-axis represents fluorescence signals at 450nm for turquoise (XBP1/IRE1 reporter). (B) Representative bi-exponential displayed dot plot of flow cytometry data in CHO CHOP::GFP and XBP1::Turquoise dual-reporter wildtype, *Eif2ak4* Δ , and *Ppp1r15a* Δ cells under 18-hour treatment of 30 μ M compound 18, 20 μ M compound 20, and 7.5 μ M compound 21 alone (magenta) or in combination with 0.75mM histidinol (yellow) (n=6). Histidinol alone is shown in blue and untreated in grey. The x-axis represents fluorescence signals at 530nm for GFP (CHOP reporter) and y-axis represents fluorescence signals at 450nm for turquoise (XBP1/IRE1 reporter). Structures of compounds 18, 20, and 21 shown on the right. (C) Quantification of B (n=6); Two-way ANOVA with Tukey's multiple comparisons test was performed and the CHOP-GFP signals (530nm) in different treatment groups were compared. *: p $\leq$ 0.05, **: p $\leq$ 0.01, ***: p $\leq$ 0.001, ****: p $\leq$ 0.0001. (D) Box and whisker plot displaying mRNA fold change of ISR effector CHOP and PPP1R15A after 4-hour treatment of 30 μ M compound 18, 10 μ M compound 20, and 10 μ M compound 21 as detected by qPCR (n=3). $\pm$ SEM, Two-way ANOVA was performed, * p $\leq$ 0.05, ** p $\leq$ 0.01.

containing a furan and two phenyl rings; compound 20, 4'-methyl-N~2~-2-pyrimidinyl-4,5'-bi-1,3-thiazole-2,2'-diamine, containing two aminothiazoles; and compound 21, 5-(4-bromophenyl)-2-(methylsulfonyl)pyrimidine (**Figure 2B** [↗](#)).

Larger quantities of compound 18 were then purchased from TimTec, USA (compound ST51216373); compound 20 was synthesised by ChemBridge, USA (compound 5935953); and compound 21 was purchased from Specs, Netherlands (compound AC-907/25005415). After validating their effects using the assays described above, we next confirmed activation of the ISR by these compounds at the transcriptional level. CHO cells were treated with 30μM of compound 18 or 10μM of compound 20 or 21 for 4 hours and expression of the ISR target genes *Chop* and *Ppp1r15a* was measured using qPCR (**Figure 2D** [↗](#)). All three induced ISR gene expression, with compound 20 eliciting the strongest response.

Cell line differences in GCN2 activator potencies

Neratinib is an ATP-competitive pan-HER kinase inhibitor used clinically in the treatment of breast cancer ([36](#) [↗](#)). Recent studies have shown that the sensitivity of glioblastoma cells to neratinib is dependent on GCN2, with GCN2 activation observed at submicromolar concentrations but inhibition at higher concentrations ([26](#) [↗](#)). We reproduced this biphasic activation of the ISR reporter in human cervical carcinoma HeLa cells and hamster CHO cells, recording EC₅₀ values for activation of 269nM (95% CI 218-322nM) and 690nM (95% CI 634-772nM) respectively (**Figure 3A** [↗](#)). Disruption of the *EIF2AK4* gene by CRISPR/Cas9 (*EIF2AK4*^{-/-}) in HeLa cells abrogated ISR activation confirming the effect to be mediated by GCN2 (**Figure 3A** [↗](#), middle panel).

In CHO cells, compound 18 was the least potent with an EC₅₀ of greater than 30μM in CHO cells without obvious inhibition up to 100μM (**Figure 3B** [↗](#)). Notably, compound 18 failed to activate the ISR measurably in either HeLa cells or human embryonic kidney 293T cells (**Figure 3C** [↗](#)). Compound 20 activated the ISR in CHO cells in a biphasic manner with an EC₅₀ of 4.8μM (95% CI 4.0-5.6μM) (**Figure 3B** [↗](#)). It was active in 293T cells though with a lower potency, EC₅₀ of 17μM, but was inactive in HeLa cells (**Figure 3C** [↗](#)). Compound 21 also showed biphasic activation of the ISR in CHO cells with an EC₅₀ of 5.0μM (95% CI 3.9-6.5μM) (**Figure 3B** [↗](#)). Weak activation was observed in 293T cells, but none was seen in HeLa cells (**Figure 3C** [↗](#)).

GCN2iB, a potent inhibitor of GCN2, abolished histidinol-induced activation of the ATF4::NanoLuc ISR reporter in CHO cells confirming this to be GCN2 mediated (**Figure 3D** [↗](#)). Supporting a role for GCN2 in mediating the compounds' effects, GCN2iB substantially reduced ISR reporter expression by each, though this effect was only partial at concentrations of compound 20 above 10μM (**Figure 3D** [↗](#)). This might indicate GCN2-independent effects or an ability to partially outcompete GCN2iB for binding at higher concentrations.

Compound 20 activates GCN2 in a GCN1-independent manner

GCN1 mediates the interaction between stalled ribosomes and GCN2 ([21](#) [↗](#), [37](#) [↗](#)). Histidinol is a competitive inhibitor of the histidyl-tRNA synthetase that depletes the cell of charged histidyl-tRNA, thus causing ribosome stalling and activation of the ISR in a GCN1-dependent manner ([37](#) [↗](#)–[39](#) [↗](#)). Using GCN1 deficient human 293T cells transiently expressing the ATF4::NanoLuc reporter, we confirmed that activation of the ISR by histidinol requires GCN1 (**Figure 4A** [↗](#)). Low concentrations of GCN2iB (5-70nM) had minimal effect on ISR reporter activation, while neratinib robustly triggered the ISR. The stimulatory effects of neratinib were significantly attenuated in GCN1 deficient cells consistent with the observation that GCN1 knockout renders astrocytoma cells resistant to neratinib ([26](#) [↗](#)). By contrast, activation of the ISR by compound 20, chosen for having the largest effect size in 293T cells, did not require GCN1 to activate the ISR reporter (**Figure 4A** [↗](#)). Indeed, loss of GCN1 modestly increased ISR activation at the highest concentration of compound 20 suggesting a mode of action independent of ribosome stalling. To gain confidence in this

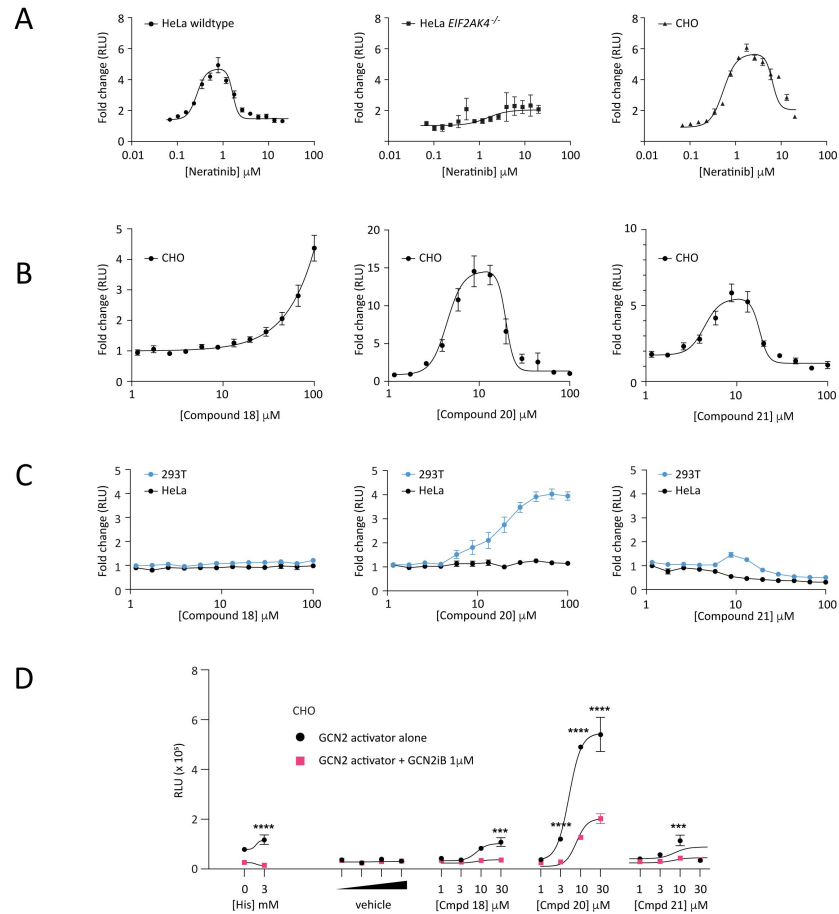


Figure 3.

Compound 18, 20, and 21 display cell type-specific differences in ISR activation.

(A) Normalised dose-dependent fold change in ATF4 expression by neratinib in HeLa WT, HeLa *EIF2AK4* Δ , and CHO WT ATF4::NanoLuc reporter cells after 16-hour treatment ($n=3$; mean $\pm$ SEM). DMSO was used as vehicle. (B) Normalised dose-dependent fold change in ATF4 expression by compound 18, 20, and 21 in CHO WT ATF4::NanoLuc reporter cells after 16-hour treatment ($n=3$; mean $\pm$ SEM). EC_{50} was estimated using 4-parameter non-linear regression. DMSO treatment was used as vehicle. (C) Normalised dose-dependent fold change in ATF4 expression by compound 18, 20, and 21 in HeLa (black) and 293T (blue) WT cells transiently transfected with ATF4::NanoLuc reporter after 16-hour treatment ($n=3$, $\pm$ SEM). DMSO treatment was used as vehicle. (D) Relative Light Unit (RLU) of ATF4 signal in CHO WT ATF4::NanoLuc reporter cells treated with 1, 3, 10, 30 μ M Compound 18, 20, and 21 alone (black) or in combination with 1 μ M GCN2iB (pink) for 18 hours ($n=3$; $\pm$ SEM). 0.1, 0.3, 1, 3% DMSO served as vehicle and 3mM histidinol as positive control. Two-way ANOVA was performed *** $p \leq 0.001$, **** $p \leq 0.0001$.

observation, we inactivated the *Gcn1* gene in CHO cells by CRISPR/Cas9. Neratinib-induced activation of the ATF4::NanoLuc ISR reporter was lost in GCN1 deficient CHO cells; however, ISR activation by compound 20 was completely insensitive to loss of GCN1 (**Figure 4B** [↗](#)).

No systematic assessment of GCN1 dependence has previously been performed for GCN2 activating drugs. We therefore proceeded to test a panel of ten additional kinase inhibitors that had been reported to activate GCN2 ([9](#) [↗](#), [11](#) [↗](#), [26](#) [↗](#)). Our panel included examples of multi-targeted tyrosine kinase inhibitors (dovitinib, sunitinib), RAF inhibitors (LY3009120, dabrafenib), inhibitors of the Wee1 kinase involved in G2 cell cycle checkpoint (WEE1-IN-4, AZD1775, Debio 0123), an inhibitor of the Heat Shock Factor 1 pathway (NXP800), and EGFR inhibitors (erlotinib and gefitinib) ([9](#) [↗](#), [11](#) [↗](#), [26](#) [↗](#), [31](#) [↗](#), [40](#) [↗](#)–[42](#) [↗](#)). In all cases, a marked reduction in GCN2 activation by the drugs was observed in GCN1 deficient cells (**Figure 4C** [↗](#)). Most of these molecules failed to activate the ISR in the absence of GCN1 (sunitinib, NXP800, dovitinib, AZD1775, dabrafenib, gefitinib, LY3009120), while for three drugs weak activation of the ISR was seen at higher concentrations (WEE1-IN-4, Debio 0123, erlotinib). None showed the dramatic GCN1 independence observed for compound 20 in unstressed cells.

ATP-competitive interaction with the GCN2 kinase domain

Existing activators of GCN2 have been shown to interact with the ATP binding pocket ([9](#) [↗](#), [11](#) [↗](#), [26](#) [↗](#)). We therefore used a bioluminescence resonance energy transfer (BRET)-based assay to examine the interaction of the three lead compounds with the kinase domain of GCN2 in cells. Briefly, the N-terminal portion of human GCN2 kinase domain fused to NanoLuc luciferase was expressed in either CHO or 293T cells. These were then treated with an ATP-binding pocket tracer. In this system, energy transfer is proportional to tracer binding to the NanoLuc-tagged kinase domain, and so interactions of other molecules that displace the tracer reduce the BRET ratio (**Figure 5A** [↗](#)) ([43](#) [↗](#)). In both CHO and 293T cells, GCN2iB efficiently suppressed the BRET ratio thus validating the system (**Figure 5B** [↗](#)). All three lead compounds significantly reduced the BRET ratio, though unlike GCN2iB, compounds 18 and 20 were noticeably more effective in CHO than in 293T cells. Although compound 21 suppressed BRET in both cell types, its effect was less dramatic in CHO cells compared to compounds 18 and 20.

To examine further the interaction of compounds 18 and 20 with GCN2's kinase domain, we generated in bacteria a recombinant protein comprising Glutathione S-transferases (GST) fused to the kinase domain of human GCN2 (residues 585 to 1005 of NCBI reference Sequence NP_001013725.2). GST was chosen both to facilitate protein purification, and to enhance dimerisation of the kinase domain in order to cause autophosphorylation and constitutive activation. When incubated *in vitro* with increasing concentrations of ATP and a fixed quantity of His6-tagged bacterially expressed eIF2 α N-terminal domain ([44](#) [↗](#)), GST-GCN2 kinase domain phosphorylated its substrate on serine 51, as detected by immunoblotting (**Figure 5C** [↗](#)). Neratinib dose-dependently inhibited eIF2 α phosphorylation (**Figure 5D** [↗](#), lanes 1-5). The lack of activation by neratinib likely reflected the requirement for autoinhibitory domains of GCN2 to observe this effect. Therefore, in this system inhibition of eIF2 α kinase activity was used as a surrogate for target engagement. Compounds 18 and 20 also inhibited eIF2 α phosphorylation, while compound 21 appeared inactive in this assay (**Figure 5D** [↗](#)). High concentrations of ATP were unable to completely overcome the inhibition by compounds 18 or 20, suggesting a mode of action, at least partially, non-competitive in nature (**Figure 5E** [↗](#) & **F**).

To test the ability of compound 20 to activate full-length GCN2, we next generated full-length human GCN2 (uniprot ID: Q9P2K8) through baculoviral expression using an N-terminal twin StrepII tag. Despite exerting only a modest effect on GCN2 autophosphorylation at T899 within the activation loop, compound 20 robustly triggered the phosphorylation of full-length eIF2 α to an even greater extent than purified tRNAs, a known endogenous stimulus of GCN2 activity (**Figure 5G** [↗](#)).

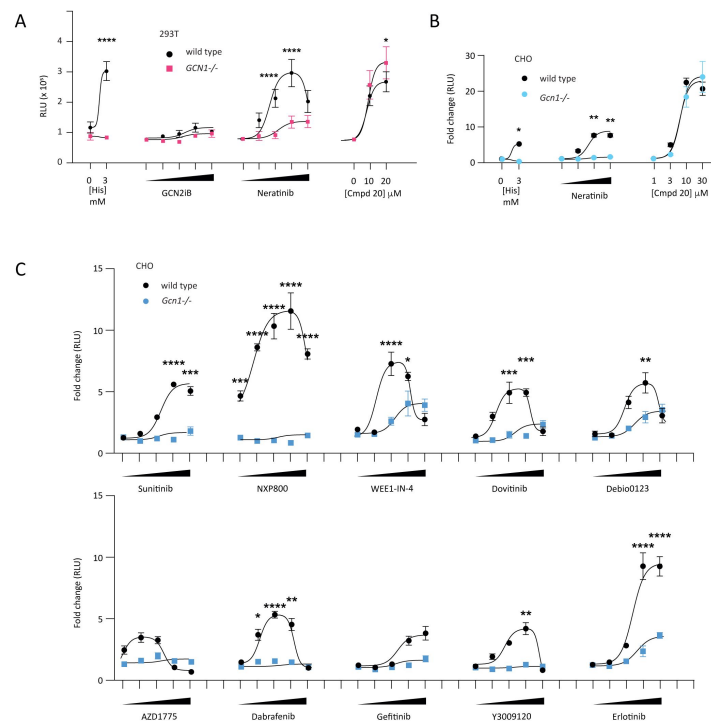


Figure 4.

Compound 20 displays GCN1 dependence.

(A) Relative Light Unit (RLU) of ATF4 signal in 293T WT (black) and GCN1Δ (magenta) cells transiently transfected with ATF4::NanoLuc reporter for 16-hour treatment with compound 20 (10 and 20 μM), GCN2iB (5, 10, 35, 75 nM), and neratinib (75, 200, 500, 1000 nM) (n=3; ± SEM). Two-way ANOVA was performed * p≤0.05, ** p≤0.01, *** p≤0.001, **** p≤0.0001. (B) Normalised fold-change in ATF4 signal in CHO WT (black) and *Gcn1*^{-/-} (blue) ATF4::NanoLuc reporter cells for 18-hour treatment with compound 20 (1, 3, 10 and 30 μM) and neratinib (75, 200, 500 nM) (n=3; ± SEM). Two-way ANOVA * p≤0.05, ** p≤0.01. (C) Normalised fold-change in ATF4 signal in CHO WT (black) and *Gcn1*^{-/-} (blue) ATF4::NanoLuc reporter cells treated for 19-hours with a panel of ATP-competitive kinase inhibitors reported to activate GCN2 (0.1, 0.3, 1, 3, and 10 μM) (n=3-9; mean ± SEM). Two-way ANOVA was performed * p≤0.05, ** p≤0.01, *** p≤0.001, **** p≤0.0001.

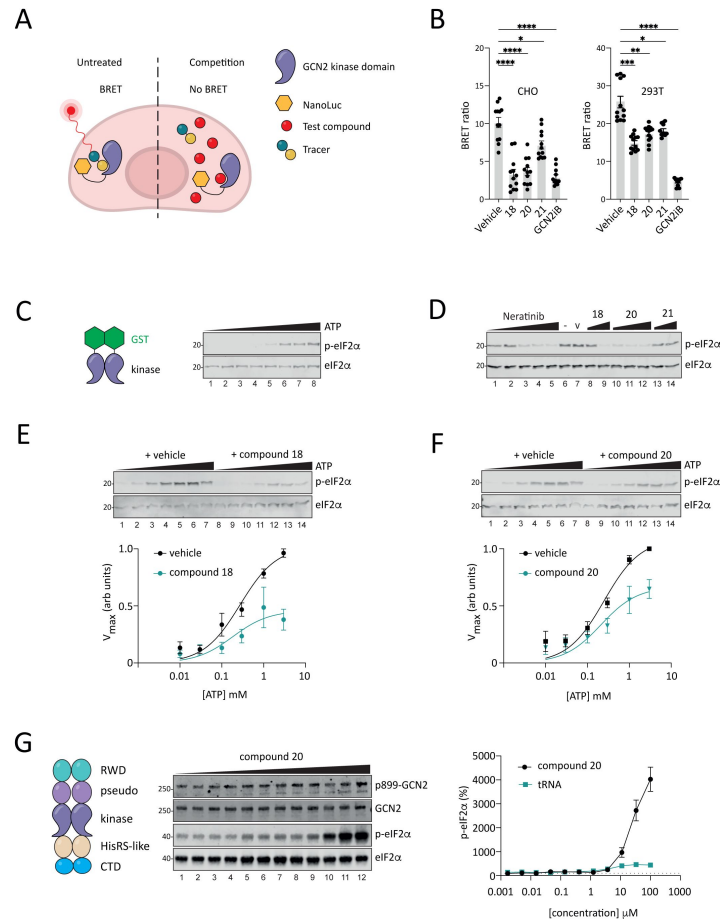


Figure 5.

Compound 18 and 20 directly binds to GCN2 in cellulo and in vitro

(A) Illustration of the NanoBRET assay. Left panel displays BRET signal at baseline with tracer binding to GCN2; right panel displays reduced BRET signal due to the addition of a competitive compound that outcompetes the tracer from binding to GCN2. (B) BRET ratio in CHO and 293T WT cells with 2-hour treatment of 3% DMSO, 100 μ M compound 18, 20, and 21 or 0.5 μ M GCN2iB. (n=3; $\pm$ SEM). Two-way ANOVA was performed. *: $p \leq 0.05$, **: $p \leq 0.01$, ***: $p \leq 0.001$, ****: $p \leq 0.0001$. (C) Schematic of GST-GCN2 fusion protein (residues 585–1005) and representative immunoblot of *in vitro* kinase assay. Purified GST-GCN2 was incubated with eIF2 α -NTD (residues 2–187) using a ramp of ATP concentrations (0, 0.01, 0.03, 0.1, 0.3, 1, 3, 10mM) (n=3). Total and serine51 phosphorylated eIF2 α were detected. (D) Representative immunoblot of *in vitro* kinase assay showing inhibition of eIF2 α -NTD (residues 2–187) phosphorylation. Purified GST-GCN2 and eIF2 α -NTD was incubated with neratinib (0.075, 0.68, 6.67, 10, 20 μ M) and compound 18 (100–500 μ M), compound 20 (100, 200, 500 μ M), and compound 21 (100–500 μ M) (n=3). 5% DMSO used as vehicle control. Total and serine51 phosphorylated eIF2 α were detected. Molecular size in kDa. (E) Representative immunoblot of *in vitro* kinase assay (n=3) showing inhibition of eIF2 α -NTD (residues 2–187) phosphorylation by 300 μ M compound 18 using a ramp of ATP concentrations (0.01, 0.03, 0.1, 0.3, 1, 3, 10mM); 3% DMSO used as vehicle control. Bottom panel showed Michaelis-Menten kinetics analysis of DMSO control (black) vs compound 18 (green). Molecular size in kDa. (F) Representative immunoblot of *in vitro* kinase assay (n=3) showing inhibition of eIF2 α -NTD (residues 2–187) phosphorylation by 100 μ M compound 20 using a ramp of ATP concentrations (0.01, 0.03, 0.1, 0.3, 1, 3, 10mM); 3% DMSO used as vehicle control. Bottom panel showed Michaelis-Menten kinetics analysis of DMSO control (black) vs compound 20 (green). Molecular size in kDa. (G) Schematic of full-length GCN2 protein and representative immunoblot of *in vitro* kinase assay. Purified full-length GCN2 was incubated with full-length eIF2 α using a ramp of compound 20 or tRNA concentrations (0, 0.001, 0.005, 0.01, 0.05, 0.13, 0.4, 1.2, 3.7, 11, 33, 100 μ M) (n=3). Total and serine51 phosphorylated eIF2 α , as well as total and threonine899 phosphorylated GCN2 were detected. On the right showed quantification of serine51 phosphorylated eIF2 α with a ramp of tRNA (black) vs compound 20 (green). Molecular size in kDa.

Interaction with other ISR kinases

The eIF2 α kinases have a common evolutionary origin and consequently the kinase domains share some structural similarity (45). We therefore examined the effect of the compounds on all four eIF2 α kinases *in vitro*. Lower concentrations of ATP (10 μ M for GCN2, 0.1 μ M for PERK, 0.15 μ M for HRI, 0.1 μ M for PKR) were used to accentuate antagonism at the ATP site. Under these conditions, in contrast to **Figure 5G** where 100 μ M ATP was used, activation of GCN2 is not observed with any of the compounds tested. At low ATP, full length, purified GCN2 was efficiently inhibited by GCN2iB with an IC₅₀ on 1.55 nM.

None of the screen hits reached 50% inhibition at the highest concentration tested (3 μ M) though compound 20 and to a lesser extent 18, appear to cause weak inhibition (**Figure S1A**). No inhibition was observed with the test compounds against purified PERK (**Figure S1B**). The PERK inhibitor GSK2606414 served as a positive control with an expected IC₅₀ of 0.66 nM. Weak inhibition of HRI by compounds 18 and 20 was observed but failed to reach 50% (**Figure S1C**). In this system, GCN2iB was found to inhibit purified HRI with an IC₅₀ of 2.76 nM, similar to that observed against GCN2. By contrast, both compounds 18 and 20 inhibited PKR *in vitro*: compound 18 with an IC₅₀ of 4.50 μ M and compound 20 with an IC₅₀ of 3.75 μ M (**Figure S1B-D**). GCN2iB inhibited PKR with an IC₅₀ of 3.26 μ M. Compound 21 was inactive against each of the four eIF2 α kinases in this assay.

In silico docking of compounds 18 and 20 to the GCN2 kinase domain and structure-activity relationship (SAR) analysis

We used the SiteMap module from Schrodinger Suite to identify binding sites based on drug interactions with nearby GCN2 residues. Published crystal structures of the GCN2 kinase domain (see methods) show the ATP binding pocket comprises a large solvent exposed region and a buried hydrophobic region that extends beyond the gatekeeper residue (Met802) and towards the IZC-helix (**Figure 6**). Compound 18 was predicted to occupy a similar pocket to that of the ligand in the PDB 6N3N crystal structure demonstrating a Type 1.5 binding configuration, characterised by the DFG loop being “in” and the IZC-helix being in the “out” conformation (10). The furan motif of compound 18 is predicted to be buried in the allosteric pocket around α C-helix, between gatekeeper Met802 and Leu640 (**Figure 6D**). The compound’s sulfone was predicted to interact with Phe867 of the conserved DFG-motif. In neratinib, a triad of hydrogen bonding was predicted between the 8-position of the isoquinoline core (Ar-H bond), the amine linker (H bond) and the 2-position of the fluorobenzene ring (Ar-H Bond) with Asp866, similar to the predicted interaction between the amide linkage of compound 18 and Asp866 (**Figure 6B**). Both neratinib and compound 18 were also predicted to protrude into the solvent exposed space (**Figure 6B-E**). However, the interaction of neratinib with Cys805 of the hinge region through the nitrile of isoquinoline core was not present in compound 18.

Due to the metabolic liabilities of furans in medicinal chemistry programs, we sought to explore the importance of this motif for binding and activity using available structural analogues. Replacement of the sulfonamide furan motif with some benzene derivatives including phenolic (BCC0106222), 2-bromobenzene (BCC0113320), 3-acetylbenzene (BCC0104214), preserved ISR activation while others (BCC0073242 and BCC0043997) resulted in a loss of activity (Supplementary Table 1). Incorporation of heterocyclic ring systems (BCC0073142, Z56783036) or modification of solvent exposed region (Z88478561, Z226209130, Z11741677) were similarly not tolerated (Supplementary Table 1). However, modifying the furan motif to benzene derivative 2-bromobenzene (BCC0113320) and heterocyclic ring (BCC0077119) preserved GCN2-specific ISR activation.

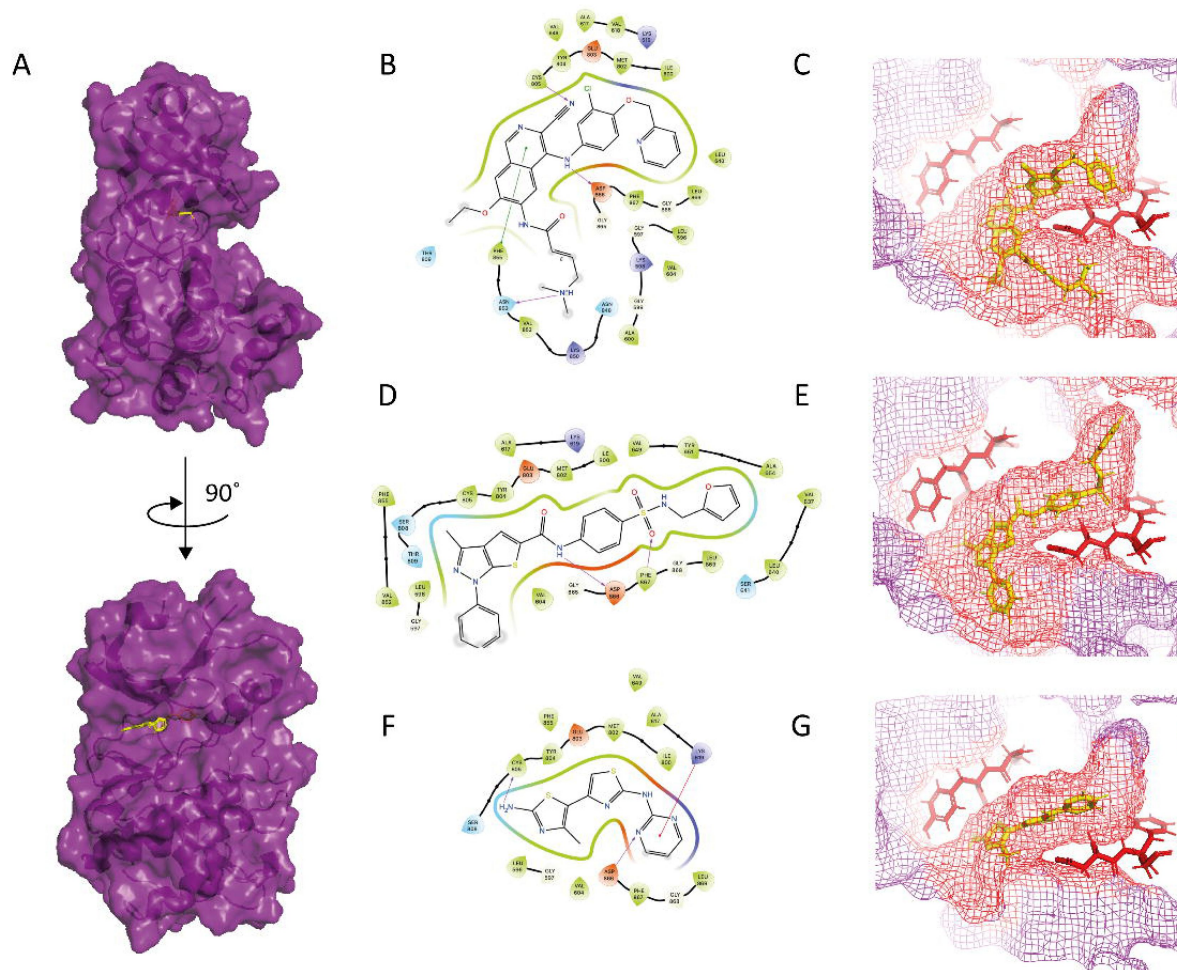


Figure 6.

Structural modelling

(A) Illustration of GCN2 kinase domain from up (N-terminus) to down (C-terminus) with illustrative ligand (yellow) in ATP-binding pocket. (B) 2D ligand interaction diagram of neratinib, (D) compound 18, and (F) compound 20. Violet, green, and red colored lining indicate hydrogen bond, π - π , and cation- π interactions respectively. Polar and hydrophobic residues are highlighted in blue and green respectively. Mesh representation of the ATP-binding pocket in GCN2 kinase domain was shown with ligand (yellow): (C) neratinib, (E) compound 18, and (G) compound 20. Key residues surrounding the ATP-binding pocket, including the DFG motif (D866-G868) and EYC residues (E803-C805) following the gatekeeper M802 were labelled red. Figures adapted from crystal structure 6N3N.

Unlike compound 18 and neratinib, compound 20 was not predicted to protrude into the allosteric pocket (**Figure 6F** & **G**). Analogues extending further towards the IZC-helix and protruding into the solvent exposed region were inactive (e.g., Z134826202 and Z118606822), suggesting that the small size of compound 20 may be advantageous, fitting into the ATP-binding pocket of GCN2 and facilitating stronger activation (Supplementary Table 1). Hydrogen bond interaction was predicted between the NH₂ of the compound 20 aminothiazole with the carbonyl of the hinge region Cys805. Additional interaction was predicted between the linker NH of compound 20 and backbone carbonyl of Asp866 in the DFG-motif. Such 2-aminopyrimidine rings are present in many kinase ligands (46). Rather than this interacting with hinge region residues, this moiety in compound 20 was predicted to form a pi-cation interaction with Lys619 and a hydrogen bond with DFG Asp866, as has been reported for other ligands of kinase ATP pockets (10). Replacement of this pyrimidine ring with 2-pyridine analogues (BCC0084738, BCC0079350 and BCC0083923) was well tolerated, likely due to the conserved hydrogen bonding interaction between the pyridine nitrogen and Asp866. Removal of a nitrogen from the ring system, such as BCC0057014, BCC0089519, BCC0084744, and BCC0084833 all resulted in a loss of activity, possibly due to their lack of interaction with Asp866 (Supplementary Table 1). Due to toxicity issues surrounding 3-aminothiazoles, modifications were made to replace the terminal thiazole ring system with phenyl derivatives (BCC0080152, BCC0075509, BCC0084830, BCC0057431 and BCC0080200).

However, these resulted in a decrease in activity towards GCN2, suggesting that there might be a steric component contributing to activity. Replacing the thiazole with a minimally steric affecting thiophene (BCC0079876) also led to activity loss, possibly due to the twisting of central thiazole ring that impaired hydrogen bonding between pyrimidine nitrogen and Asp866. We were therefore unable to improve upon compound 20's activity but gained valuable SAR information for future development.

Discussion

The ISR plays a critical role for maintaining homeostasis under diverse stress conditions and its disruption results in a variety of disease states (2). The observation that some pathogenic variants of GCN2 are amenable to pharmacological activation led us to seek novel GCN2 activators. Our chemical screen identified three chemically distinct GCN2 activating molecules, one of which is, to our knowledge, the first to show independence from GCN1 for its activity. This may prove useful as a tool compound for the study of GCN2 in cells and using purified components since it overcomes at least one hurdle to kinase activation.

Compounds 18 and 20 demonstrate weak inhibition of the related kinase PKR. Previous structural analyses have shown the kinase domains of PKR and GCN2 to be highly conserved, particularly at the ATP-binding pocket, which likely contributes to this effect. Indeed, a previous publication reported that other small molecule inhibitors of GCN2, including a series based on a triazolo[4,5-d]pyrimidine scaffold, demonstrated cross-inhibition of PKR (47). Furthermore, the PKR inhibitor C16 has been shown to activate GCN2 *in vitro* (11). Intriguingly, viral infections or synthetic dsRNA known to activate PKR, have also been shown to trigger GCN2 in some circumstances, though if this relates to structural similarities of their kinase domains has not been studied (11, 48, 49).

A remarkable finding in the current study was the lack of dependence on GCN1 for compound 20 to activate GCN2 in cells. During amino acid starvation, ribosome-associated GCN1 facilitates binding of uncharged tRNAs to GCN2, thus facilitating GCN2 activation (21, 50, 51). GCN1 also functions as a disome detector, enabling GCN2 activation in response to ribosome collisions (37). In genetic screens identifying modifiers of GCN2 activation by a small molecule, such as the dual HER2/EGFR inhibitor neratinib and the WEE1 inhibitor AZD1775, GCN1 has repeatedly been found to be necessary (26, 30, 31). We were able to confirm this GCN1 dependence for

neratinib in the present study. It was therefore surprising that GCN2 activation by compound 20 did not appear to require GCN1 in human or hamster cells. While GCN1 is essential for the activation of GCN2 by amino acid starvation in yeast (52 [↗](#), 53 [↗](#)), *in vitro* studies have shown that GCN2 activation by the ribosome P-stalk or by deacylated tRNAs does not absolutely require GCN1 (24 [↗](#), 25 [↗](#)). Yet we found compound 20 able to activate purified full-length human GCN2 to a far higher level than was possible with purified tRNAs. Our modelling suggests that unlike neratinib, compound 20 can fit within the ATP-pocket without needing to protrude into the solvent, nor extending deeply into the hydrophobic region beyond the gatekeeper Met802 residue towards the IZC-helix. Indeed, analogues extending in these directions were inactive, perhaps suggesting value to this molecule's small size. This may prove useful as a tool compound for studies in which GCN1 is unavailable or non-functional.

There are limitations to our study. So far, our experiments have been limited to cultured cells, purified proteins, and *in silico* modelling. We have yet to test our compounds in an animal model. Although our pipeline excluded molecules showing toxicity, it is likely that prolonged activation of the ISR will have toxic consequences via targets including CHOP and PPP1R15A (16 [↗](#)). Our most promising molecule, compound 20, contains a 2-aminothiazole. Although known to be a versatile scaffold, appearing in several clinically useful drugs including anti-inflammatory, anti-microbial and anticancer agents, 2-aminothiazole is sometimes considered a toxicophore owing to toxic consequences following its metabolism, and as a pan-assay interference compound (PAINS), owing to frequent false positives in high-throughput drug screens (54 [↗](#)–56 [↗](#)). Our use of orthogonal approaches to support its activity as a GCN2 activator, mitigate against it being a non-specific hit. Nevertheless, if this compound were to be developed further, attempts to ‘design out’ this feature by, for example, replacing the sulphur for an oxygen (thiazole to oxazole switch) could be attempted. Indeed, computational modelling reveals an adjacent NH which should form a stronger H-bond to oxygen of an oxazole, which may result in an improved potency to GCN2. Another potential limitation of our study is the use of existing analogues in SAR studies, the so-called ‘analogue by catalogue’ approach. Although less systematic than chemically synthesising further series *de novo*, it was cost effective and generated useful information on the tolerability of modifications of our lead scaffolds. Finally, our compounds were identified in a screen of CHO cells and appear to be more effective in these cells than in HeLa and 293T models. The absence of detectable activity in HeLa cells does not appear to be a species effect, since some activity was observed in human 293T cells and when using recombinant human GCN2 protein. It is likely therefore to reflect cell-type effects, since cancer cell lines are known to demonstrate dramatic differences in drug efficiency, even when using different strains of the same cell line (57 [↗](#)). This should be considered carefully when using these compounds as tools or for further drug development.

In summary, we performed a chemical screen for novel activators of GCN2. We identified three chemically distinct scaffolds, and demonstrated one to be independent of the GCN2 cofactor GCN1.

Methods

Screen molecules

The initial screen used compounds sourced from the BioAscent library and all molecules dissolved in DMSO. Some follow on compounds for SAR were sourced from Enamine. Subsequently, compound 18 was purchased from TimTec, USA (compound ST51216373). Compound 20 was synthesised by ChemBridge, USA (compound 5935953). Compound 21 was ordered from Specs, Netherlands (compound AC-907/25005415).

NanoGlo® luciferase assay

pGL4.2 ATF4-Nluc and pGL4.2 ATF4-Nluc-PEST® were described previously (17). In 384-well plate, test compounds, 3mM histidinol or vehicle control were plated as 5x solution in a volume of 5µL in 10% FBS-Nutrient Mixture F-12 Ham supplemented with 1x GlutaMAX™. CHO wildtype and *Eif2ak4*^{-/-} ATF4::NanoLuc reporter cells were subsequently seeded at a density of 1.5×10^3 cells per well in 20µL. After 16 hours at 37°C, NanoGlo® luciferase assay reagent was added to each well and mixed with orbital shaking for 1 minute. After 10 minutes of incubation at room temperature, bioluminescence signal (360-545 nm) was acquired on a Tecan Spark plate reader. Cells carrying a ATF4::nanoLuc-PEST reporter were used for live-cell assay, allowing detection of time-sensitive ATF4 signals (17). Cells (1.25×10^4 cells per well) were plated overnight on white, clear bottom 96-well plate. Next day, 1µL 100x Nano-Glo® Endurazine™ substrate was mixed with cell culture medium and added to cells and incubated for 2 hours to reach stabilisation. Compounds were added to the wells and luminescence signals were measured using a Tecan CM Spark plate reader at 37°C in a humidity cassette for 48 hours. Data analysed using Graphpad Prism 10.

Bioluminescence Resonance Energy Transfer (BRET)

NanoBRET assay was performed according to user's manual. Briefly, 4×10^5 cells/mL (HEK293T or CHO) were transfected with 100ng of GCN2 kinase domain::nanoLuc reporter and 900ng of transfection carrier DNA mixed in 100µL Opti-mem (no phenol red) using lipofectamine LTX, and plated in 384-well white plate (40µL/well, 8×10^3 cells). Twenty-two hours after transfection, tracer was added to each well and mixed on an orbital shaker. Test compound (final concentration 10, 30, 100µM), DMSO control (final concentration 1, 3, 10%), and positive control GCN2iB (final concentration 1µM) were added to each well and mixed. After 2 hours of incubation, NanoBRET™ Nano-Glo® substrate-extracellular NanoLuc® Inhibitor reagent was added to each well before bioluminescence measurement using a Tecan CM Spark plate reader (multi-colour measurement mode: wavelength 1, wavelength 445-470nm, 300ms integration; wavelength 2, wavelength 610-635nm, 300ms integration). Data analysed using Graphpad Prism 10.

Cell Viability Assay

Compounds were plated into 384-well white plates and mixed with 20µL of culture medium containing 1.5×10^3 cells per well, incubated for 16 hours at 37°C. CellTiter-Glo® Assay reagent was added to each well for bioluminescence measurement using a Tecan CM Spark plate reader. Data analysed using Graphpad Prism 10.

Flow cytometry

CHOP::GFP and XBP1::Turquoise dual-reporter cells were treated with each compound (7.5-20µM) and histidinol (0.5mM) 18-20 hours prior to cell harvesting (35). ER stress was induced using 2µg/mL of tunicamycin. At the time of experiment, cells were trypsinised and collected by centrifugation, resuspended in 4mM EDTA PBS, filtered through a 50µm filter, then analysed using a DB Fortessa flow cytometer. GFP was measured at 530nm and Turquoise was measured at 450nm. Data analysed using FlowJo software.

Quantitative real time PCR (qPCR)

RNA was reverse transcribed to produce complementary DNA using High-Capacity cDNA Reverse Transcription Kit (Thermo, 4368814) according to the manufacturer's instructions. To perform realZtime PCR, the PowerUp SYBR Green Master Mix (Thermo A25742) and the CFX96 BioRad realZtime PCR machine (Biorad, UK) were used. All samples were measured in duplicate and included water control. Fold changes in expression were calculated using the $\Delta\Delta C_t$ method. Hamster qPCR primers include CHOP (Fw: GGGAGCTGGAAGCCTGGTAT, Rv:

GGGACCCCATTTTCATCTG), PPP1R15A (Fw: CCTGGTCTGCAAAGTGCTGAT, Rv: CCAGCTCAGTCACTCCCTCTTC), and β -actin (Fw: ACTCCTACGTGGGTGACGAG, Rv: AGGTGTGGTGCCAGATCTTC). Data analysed using Graphpad Prism 10.

Immunoblotting

Cells were washed with PBS on ice and lysed in low-salt buffer as previously described (17 [↗](#)). Following SDS-PAGE (7% gel for GCN2 and 14% gel for eIF2 α or alternatively, 5-12% gradient gel), proteins were transferred to nitrocellulose membranes then analysed by immunoblotting. Antibodies used: total GCN2 (Gift from Ron lab: NY168); total GCN2 (1:1000, Invitrogen JA03-83); GCN2-phosphoT899 (1:1000, Abcam, ab75836); β -actin (1:5000, Abcam, ab8226); eIF2 α phos-S51 (1:1000, Abcam, ab32157); eIF2 α phos-S51 (1:500, Cell Signalling Technology 9721); total eIF2 α (1:1000, Invitrogen, AHO0802); total eIF2 α (Gift from David Ron lab, CIMR); total eIF2 α (1:1000, Santa Cruz Biotechnology sc-133132).

Recombinant GCN2 kinase domain synthesis and in vitro kinase assay

The human GCN2 kinase domain (residues 585 to 1005) was tagged on its N-terminus with GST by cloning into the vector pGV67 by Gibson assembly. Protein was synthesised in BL21(DE3) *E. coli* Rosetta cells and purified using glutathione agarose beads. GCN2 kinase domain (0.3 μ g) was incubated with 100 μ M ATP, 1 μ M eIF2 α -NTD and test compounds in reaction buffer (50mM HEPES, pH 7.4, 100mM potassium acetate, 5mM magnesium acetate, 250 μ g/ml BSA, 10mM magnesium chloride, 5mM DTT, 5 mM β -glycerophosphate). Reactions were incubated at 32°C for 10 minutes, then immediately quenched with SDS sample buffer pre-warmed to 95°C. Samples were then analysed by immunoblotting.

Recombinant full-length GCN2 synthesis and in vitro kinase assay

Expression and purification of recombinant human GCN2 was conducted as described previously (25 [↗](#)). Briefly, human GCN2 (uniprot ID: Q9P2K8) was cloned into a baculoviral vector with an N-terminal twin StrepII tag followed by a TEV protease site. GCN2 was then expressed in *Sf9* cells grown at 27 °C for 55 hours. Protein was purified using sequential StrepTrap HP column (GE Healthcare Life Sciences 28-9075-47), HiTrap Q HP column (Cytiva 17115401), Hi Load 16/60 Superdex 200 column (Cytiva/GE Healthcare 28989335). Ten nanomolar GCN2 was incubated at 30 °C for 20 min with 2 μ M eIF2 α and 100 μ M ATP in increasing concentrations (0.00169-100 μ M) of deacylated tRNA or compound 20. Samples were then quenched via the addition of SDS-PAGE loading buffer and heating to 95 °C for 3 min, then analysed by SDS-PAGE and immunoblotting: total eIF2 α (Santa Cruz Biotechnology sc-133132) 1:1000, phospho-eIF2S1 Ser51 (Cell Signalling Technology 9721) 1:500, total GCN2 (Invitrogen JA03-83) 1:1000, pThr899 GCN2 (Abcam AB75836) 1:1000. Images were captured and quantified in Image Studio Ver 5.2. Data analysed using Graphpad Prism 10.

Expression and purification of eIF2 α

Expression and purification of recombinant human eIF2 α was conducted as described previously (25 [↗](#)). DNA encoding full-length human eIF2 α (NCBI reference number: NP_004085.1) was inserted into the vector pOPTH with an N-terminal His6 tag followed by a TEV protease site. Protein was expressed in BL21 Star (DE3) cells, at 37 °C for 3 hours then purified using a HisTrap HP Column (Cytiva 17524801) as described for GCN2.

LanthaScreen in vitro kinase activity assay (Figure S1 [↗](#))

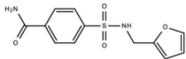
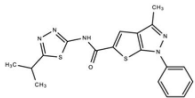
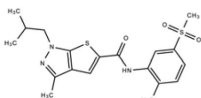
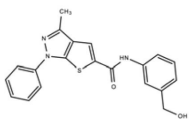
In 384-well plate format, 1nM human GCN2 (Carna Bioscience, # 05-153) or 1nM human PERK (Carna Bioscience, # 05-155) or 2nM human HRI (Carna Bioscience, # 05-154) or 3nM human PKR (Carna Bioscience, # 05-156) were incubated with 50mM HEPES pH 7.0, MgCl₂ 10mM, MnCl₂ 5mM,

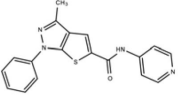
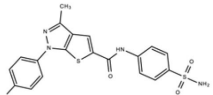
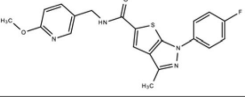
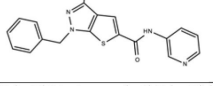
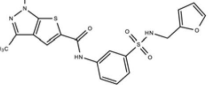
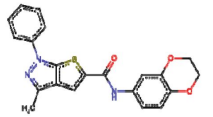
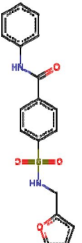
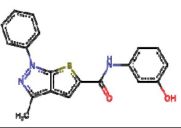
tRNA 4μM (only for GCN2 assays), ATP (10μM for GCN2, 0.1μM for PERK, 0.15μM for HRI, 0.1μM for PKR), GFP-eIF2α 80nM. Reaction time was 20-30 minutes at room temperature, followed by detection time 50-60 minutes. TR-FRET signal between terbium-labeled antibody and GFP-labeled p-eIF2α read with excitation 340nm and emission of the donor (terbium labeled anti-p-eIF2α antibody: 495 nm) and the acceptor (GFP-labeled p-eIF2α: 520nm) on the Tecan Spark plate reader. IC₅₀ values were determined in GraphPad Prism 8.0, using a 4-parameter model: log(inhibitor) vs. response – variable slope.

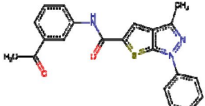
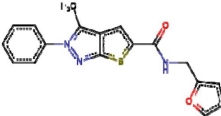
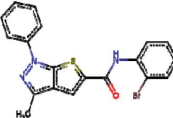
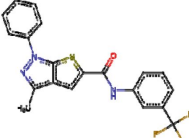
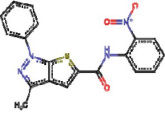
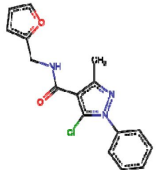
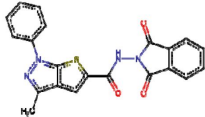
In silico modelling

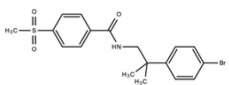
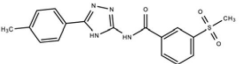
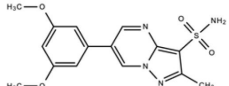
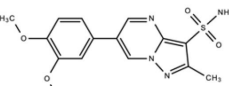
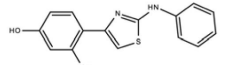
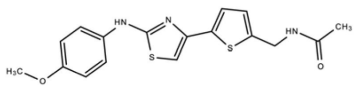
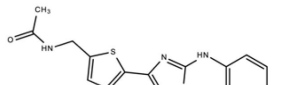
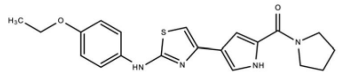
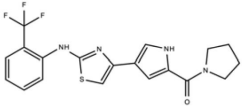
Five reported crystal structures for human GCN2 were considered (PDB 6N3N, 6N3O, 6N3L, 7QWK, and 7QQ6 (10, 58)). Of these, 6N3N and 6N3L share ~100% sequence homology. While 6N3N has unresolved amino acids residues, these are remote from the ATP binding site and should not impact the binding of the ligands. A visual inspection of 6N3O, 6N3N, and 6N3L highlighted that the activation loop in 6N3L is cut short and does not extend beyond the IZC-helix, affecting the orientation of the DFG loop and its residues (Asp866, Phe867, Gly868). As such, 6N3L was not considered for further analysis. PDB sequences 7QWK and 7QQ6 share ~75% sequence homology, with 7QWK containing unresolved residues remote from the ATP binding site. As observed with PDB 6N3L, the activation loop containing the DFG residues is cut short in both cases and does not extend beyond the IZC-helix. Inspection of the binding mode of crystallised ligands of 7QQ6 and 7QWK demonstrate that these bind in a Type I binding pocket, compared to Type II binding as observed in 6N3O.

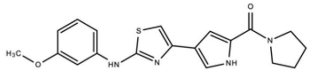
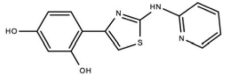
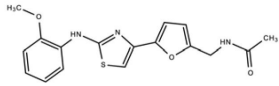
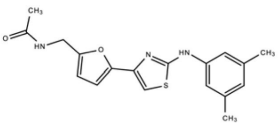
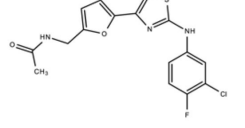
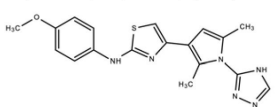
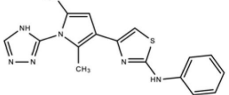
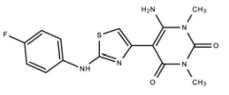
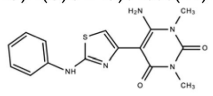
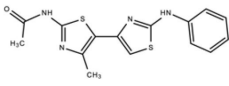
Prior to docking, the protein structure (PDB ID: 6N3N) was put through the Protein Preparation Protocol of Maestro. This protocol added hydrogens, removed co-crystallised water molecules, optimised the receptor's hydrogen bond network, enumerated bond orders, and performed a restrained minimisation to alleviate backbone clashes with the receptor backbone. The protein was prepared at pH7.4 (59). SiteMap module was used on the prepared protein, with all settings set to default (SiteMap, Schrödinger Release 2024-4, Schrödinger, LLC, New York, <http://www.schrodinger.com/>). Grid generation was performed using the Glide Docking Protocol in Masestro (Glide, Schrödinger Release 2024-4, Schrödinger, LLC, New York, <http://www.schrodinger.com/>). This was centred around Met802 and Lys619. High throughput virtual screening (HTVS) setting was used for docking.

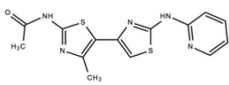
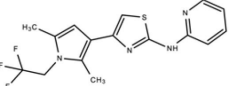
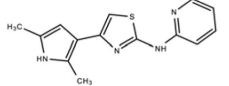
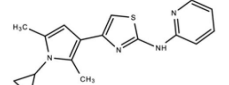
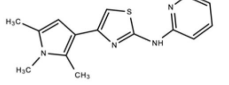
Structure of compound 18 analogue and fragment	ID	ISR activation	GCN2 specificity
<chem>NC(=O)c1ccc(cc1)S(=O)(=O)NCc1ccc(O)cc1</chem> 	Z33546257	-	NA
<chem>CC(C)c1nnc(s1)NC(=O)c1cc2c(nn(c2s1)-c1ccccc1)C</chem> 	Z56783036	-	NA
<chem>CC(C)Cn1nc(c2cc(sc12)C(=O)Nc1cc(ccc1C)S(=O)(=O)C)C</chem> 	Z88478561	-	NA
<chem>Cc1nn(c2sc(cc12)C(=O)Nc1cccc(c1)CO)-c1ccccc1</chem> 	Z28913272	+	NA
<chem>Cc1nn(c2sc(cc12)C(=O)Nc1ccncc1)-c1ccccc1</chem>	Z85517350	++	NA

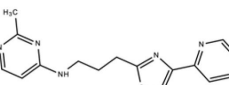
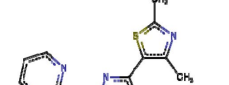
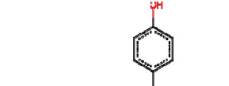
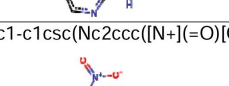
			
<chem>Cc1nn(c2sc(cc12)C(=O)Nc1ccc(cc1)S(=O)(=O)N)-c1ccc(cc1)Cl</chem>	Z27682945	+	NA
			
<chem>COc1ccc(cn1)CNC(=O)c1cc2c(nn(c2s1)-c1ccc(cc1)F)C</chem>	Z226209130	-	NA
			
<chem>Cc1nn(c2sc(cc12)C(=O)Nc1cccnc1)Cc1ccccc1</chem>	Z117416778	+	NA
			
<chem>Cc1nn(c2sc(cc12)C(=O)Nc1cccc(c1)S(=O)(=O)NCc1ccco1)C</chem>	Z228719934	-	NA
			
<chem>Cc1nn(-c2cccc2)c2sc(C(=O)Nc3ccc4c(c3)OCCO4)cc12</chem>	BCC0077119	+	+
			
<chem>O=C(Nc1cccc1)c1ccc(S(=O)(=O)NCc2ccco2)cc1</chem>	BCC0112913	-	NA
			
<chem>Cc1nn(-c2cccc2)c2sc(C(=O)Nc3cccc(O)c3)cc12</chem>	BCC0106222	++	-
			
<chem>CC(=O)c1cccc(NC(=O)c2cc3c(C)nn(-c4cccc4)c3s2)c1</chem>	BCC0104214	+	NA

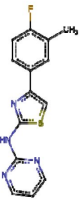
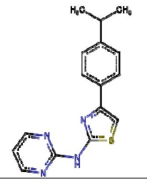
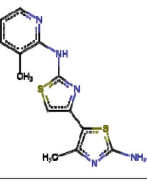
			
<chem>Cc1c2cc(C(=O)NCc3ccco3)sc2nn1-c1ccccc1</chem>	BCC0040387	-	NA
	BCC0113320	+	+
<chem>Cc1nn(-c2ccccc2)c2sc(C(=O)Nc3ccccc3Br)cc12</chem>	BCC0073142	-	NA
	BCC0043997	-	NA
<chem>Cc1nn(-c2ccccc2)c2sc(C(=O)Nc3ccccc3)cc12</chem>			
	BCC0073242	-	NA
<chem>Cc1nn(-c2ccccc2)c2sc(C(=O)Nc3ccccc3[N+](=O)[O-])cc12</chem>	BCC0104192	-	NA
	BCC0074259	-	NA
<chem>Cc1nn(-c2ccccc2)c(Cl)c1C(=O)NCc1ccco1</chem>			
			
<chem>Cc1nn(-c2ccccc2)c2sc(C(=O)NN3C(=O)c4ccccc4C3=O)cc12</chem>			
			

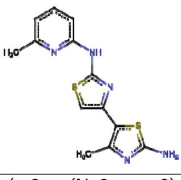
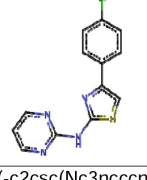
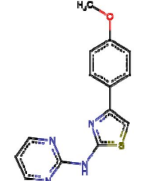
Structure of compound 21 analogue and fragment	ID	ISR activation	GCN2 specificity
<chem>CC(C)(CNC(=O)c1ccc(cc1)S(=O)(=O)c1ccc(cc1)Br</chem> 	Z423087112	-	NA
<chem>Cc1ccc(cc1)-c1nnc([nH]1)NC(=O)c1cccc(c1)S(=O)(=O)C</chem> 	Z119766318	-	NA
<chem>COc1cc(cc(c1)-c1cnc2c(c(nn2c1)C)S(=O)(=O)N)OC</chem> 	Z3464816516	-	NA
<chem>COc1ccc(cc1OC)-c1cnc2c(c(nn2c1)C)S(=O)(=O)N</chem> 	Z3464816527	-	NA
Structure of compound 20 analogue and fragment	ID	ISR activation	GCN2 specificity
<chem>Oc1ccc(c(c1)O)-c1csc(n1)Nc1ccccc1</chem>	Z48853885	-	NA
			
<chem>COc1ccc(cc1)Nc1nc(cs1)-c1ccc(s1)CNC(=O)C</chem> 	Z73485298	-	NA
<chem>CC(=O)NCc1ccc(s1)-c1csc(n1)Nc1ccc(cc1)F</chem> 	Z118606822	-	NA
<chem>CCOc1ccc(cc1)Nc1nc(cs1)-c1cnc(c1)C(=O)N1CCCC1</chem> 	Z134826202	++	NA
<chem>FC(F)(F)c1ccccc1Nc1nc(cs1)-c1cnc(c1)C(=O)N1CCCC1</chem> 	Z134826094	+	NA
<chem>COc1cccc(c1)Nc1nc(cs1)-c1cnc(c1)C(=O)N1CCCC1</chem>	Z134826982	++	NA

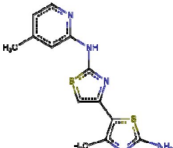
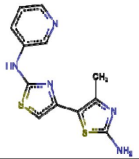
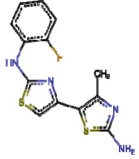
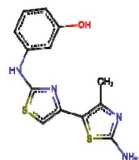
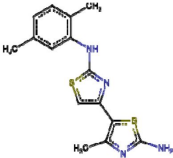
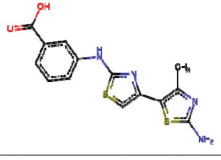
			
<chem>Oc1ccc(c(c1)O)-c1csc(n1)Nc1ccccc1</chem>	Z238766124	+	NA
			
<chem>COc1ccccc1Nc1nc(cs1)-c1ccc(o1)CNC(=O)C</chem>	Z48856426	-	NA
			
<chem>CC(=O)NCc1ccc(o1)-c1csc(n1)Nc1cc(cc(c1)C)C</chem>	Z48859481	-	NA
			
<chem>CC(=O)NCc1ccc(o1)-c1csc(n1)Nc1ccc(c(c1)Cl)F</chem>	Z48857131	-	NA
			
<chem>COc1ccc(cc1)Nc1nc(cs1)-c1cc(n(c1C)-c1nnc[nH]1)C</chem>	Z204017924	-	NA
			
<chem>Cc1cc(c(n1-c1nnc[nH]1)C)-c1csc(n1)Nc1ccccc1</chem>	Z204003168	-	NA
			
<chem>CN1C(=C(C(=O)N(C)C1=O)c1csc(n1)Nc1ccc(cc1)F)N</chem>	Z118606410	-	NA
			
<chem>CN1C(=C(C(=O)N(C)C1=O)c1csc(n1)Nc1ccccc1)N</chem>	Z94733799	-	NA
			
<chem>Br.CC(=O)Nc1nc(c(s1)-c1csc(n1)Nc1ccccc1)C</chem>	Z2854692688	-	NA
			
<chem>Br.CC(=O)Nc1nc(c(s1)-c1csc(n1)Nc1ccccc1)C</chem>	Z2239043599	-	NA

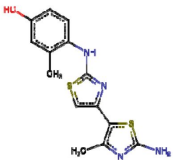
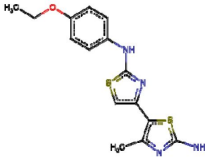
			
Cc1cc(c(n1CC(F)(F)F)C)-c1csc(n1)Nc1cccn1	Z238766162	+	NA
			
Cl.Cc1cc(c([nH]1)C)-c1csc(n1)Nc1cccn1	Z381026350	-	NA
			
Cl.Cc1cc(c(n1C1CC1)C)-c1csc(n1)Nc1cccn1	Z357935644	-	NA
			
Cl.Cc1cc(c(n1C)C)-c1csc(n1)Nc1cccn1	Z381543540	-	NA
			
Cc1nccc(n1)NCCCc1nc(cs1)-c1cccn1	Z2582963583	-	NA

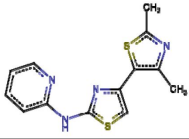
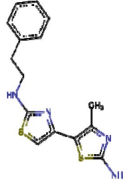
			
Cc1nc(C)c(-c2csc(Nc3ncccn3)n2)s1	BCC0083926	-	NA
			
Oc1ccc(-c2csc(Nc3ncccn3)n2)cc1	BCC0089540	+	+
			
Cc1nc(N)sc1-c1csc(Nc2ccc([N+](=O)[O-])cc2)n1	BCC0084097	++	-
			
Cc1cc(-c2csc(Nc3ncccn3)n2)ccc1F	BCC0080200	-	NA

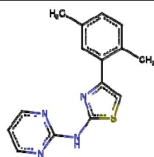
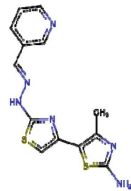
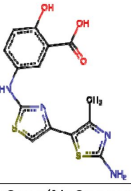
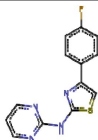
			
<chem>CC(C)c1ccc(-c2csc(Nc3ncccn3)n2)cc1</chem>	BCC0080152	-	NA
			
<chem>Cc1cccn1Nc1nc(-c2sc(N)nc2C)cs1</chem>	BCC0084738	+	+
			
<chem>Cc1cccc(Nc2nc(-c3sc(N)nc3C)cs2)n1</chem>	BCC0084842	-	NA

			
<chem>Clc1ccc(-c2csc(Nc3ncccn3)n2)cc1</chem>	BCC0075509	-	NA
			
<chem>COc1ccc(-c2csc(Nc3ncccn3)n2)cc1</chem>	BCC0057431	-	NA
			
<chem>Cc1ccnc(Nc2nc(-c3sc(N)nc3C)cs2)c1</chem>	BCC0083923	+	+

			
<chem>Cc1nc(N)sc1-c1csc(Nc2ccccc2)n1</chem>	BCC0084758	-	NA
	BCC0084762	+	-
<chem>Cc1nc(N)sc1-c1csc(Nc2ccccc2F)n1</chem>	BCC0105427	-	NA
			
<chem>Cc1nc(N)sc1-c1csc(Nc2ccccc2O)c2)n1</chem>			
			
<chem>Cc1ccc(C)c(Nc2nc(-c3sc(N)nc3C)cs2)c1</chem>	BCC0084833	-	NA
	BCC0057014	-	NA
<chem>Cc1nc(N)sc1-c1csc(Nc2ccccc2C(=O)O)c2)n1</chem>	BCC0084814	-	NA
			
<chem>Cc1cc(O)ccc1Nc1nc(-c2sc(N)nc2C)cs1</chem>			

			
<chem>CCOc1ccc(Nc2nc(-c3sc(N)nc3C)cs2)cc1</chem>	BCC0087543	+	-
			
<chem>Cc1nc(N)sc1-c1csc(Nc2cccc2[N+](=O)[O-])n1</chem>	BCC0084744	-	NA
<chem>Cc1nc(C)c(-c2csc(Nc3cccn3)n2)s1</chem>	BCC0079350	++	-

			
<chem>Cc1nc(N)sc1-c1csc(NCCc2ccccc2)n1</chem>	BCC0089615	-	NA
			
<chem>c1cnc(Nc2nc(-c3cccs3)cs2)nc1</chem>	BCC0079876	-	NA
<chem>Cc1ccc(C)c(-c2csc(Nc3ncccn3)n2)c1</chem>	BCC0080199	-	NA

			
<chem>Cc1nc(N)sc1-c1csc(N/N=C/c2cccnc2)n1</chem>	BCC0041121	-	NA
	BCC0089519	-	NA
<chem>Cc1nc(N)sc1-c1csc(Nc2ccc(O)c(C(=O)O)c2)n1</chem>	BCC0084830	-	NA
			
<chem>Fc1ccc(-c2csc(Nc3ncccn3)n2)cc1</chem>			
			
<chem>Cc1cc(C)nc(NS(=O)(=O)c2ccc(Nc3nc(-c4sc(N)nc4C)cs3)cc2)n1</chem>	BCC0087668	-	NA

NA = not applicable; assay not performed due to the inactivity of the analogue c

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Additional files

Supplementary Figure 1. Compound activity across eIF2 α kinases using LanthaScreen in vitro kinase activity assay Left panels: dose-dependent inhibition of GCN2, PERK, HRI or PKR with positive control drug GCN2iB or GSK2606414. Right panels: effect of concentration ranges of compounds 18, 20, and 21 in this system revealed weak inhibition of GCN2, HRI, and PKR. Low ATP concentrations were used to accentuate inhibition, and this masks the stimulatory effects on GCN2. [↗](#)

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Reviewer #1 (Public review):

Summary:

This manuscript describes a chemical screen for activators of the eIF2 kinase GCN2 (EIF2AK4) in the integrated stress response (ISR). Recently, reported inhibitors of GCN2 and other protein kinases have been shown at certain concentrations to paradoxically activate GCN2. The study uses CHO cells and ISR reporter screens to identify a number of GCN2 activator compounds, including a potent "compound 20." These activators have implications for the development of new therapies for ISR-related diseases. For example, although not directly pursued in this study, these GCN2 activators could be helpful for the treatment of PVOD, which is reported for patients with certain GCN2 loss-of-function mutations. The identified activators are also suggested to engage with the GCN2 directly and can function while devoid of GCN1, a co-activator of GCN2.

Strengths:

The manuscript appears to be a largely rigorous study that flows in a logical manner. The topic is interesting and significant.

Weaknesses:

Portions of the manuscript are not fully clear. There are some experimental presentation and design concerns that should be addressed to support the stated conclusions.

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Reviewer #2 (Public review):

Summary:

In this manuscript, Zhu, Emanuelli, and colleagues describe a novel pharmacological activator of the Integrated Stress Response kinase GCN2. The work is conclusive and biochemically solid. This work significantly adds to the pharmacological arsenal targeting the ISR and, in particular, GCN2.

Strengths:

Strong biochemistry, novel molecular activator of GCN2 (GCN1 independent).

Weaknesses:

The rationale for the screen is not exploited in the results (e.g., pathogenic GCN2 mutants), and lots of cell-based read-outs are not endogenous.

Major points

- (1) Regarding the justification of the work. Since the authors justify the screen for GCN2 activators with loss-of-function mutants associated with diseases, it would be of interest to evaluate whether the best compounds identified in the study are indeed able to prompt activation of those mutants (or at least of the most prevalent). This approach could actually go in parallel with the docking experiments carried out in the last figure of the manuscript, where mutants could be modeled as well.
- (2) The compounds are only tested using « artificial » proximal signaling outputs. It would be interesting to evaluate whether the best identified compounds are capable of prompting endogenous eIF2alpha phosphorylation in cellular models.
- (3) Other GCN2 activators (other than GCN2iB, e.g., HC-7366) were recently identified. In this context, it would be of interest to carry out a small benchmarking study to evaluate how the compounds identified in the current study perform against the previously identified molecules.

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Reviewer #3 (Public review):

Summary:

In this manuscript, the authors describe the results of a high-throughput screen for small-molecule activators of GCN2. Ultimately, they find 3 promising compounds. One of these three, compound 20 (C20), is of the most interest both for its potency and specificity. The major new finding is that this molecule appears to activate GCN2 independent of GCN1, which suggests that it works by a potentially novel mechanism. Biochemical analysis suggests that each binds in the ATP-binding pocket of GCN2, and that at least in vitro, C20 is a potent agonist. Structural modeling provides insight into how the three compounds might dock in the pocket and generates testable hypotheses as to why C20 perhaps acts through a different mechanism than other molecules.

Strengths:

Of the 3 compounds identified by the authors, C20 is the most interesting, not just for its intriguing mechanistic distinction as being GCN1-independent (shown genetically in two distinct cell lines, CHO and 293T in Figure 4, and in contrast to other GCN2 activators) but also for its potency. In in-cellulo assays, compound 21 appears as more of an ISR enhancer than an activator per se, and although compound 18 and compound 21 lead to upregulation of the ISR targets (Figure 2), that degree of upregulation is probably not significantly different from that induced by those compounds in Gcn2^{-/-} cells. For C20, the effect appears stronger (although it is unclear whether the authors performed statistical analysis comparing the two genotypes in Figure 2D). In Figure 3, only C20 activates the ISR robustly in both CHO and 293T. Ultimately, C20 might be a tool for providing mechanistic insight into the details of GCN2 activation and regulation, and could be exploited therapeutically.

Weaknesses:

There are some limitations to the existing work. As the authors acknowledge, they do not use any of the compounds in animals; their in vivo efficacy, toxicity, and pharmacokinetics are unknown. But even in the context of the in cellulo experiments, it is puzzling that none of the

three compounds, including C20, has any effects in HeLa cells when Neratinib does. It's beyond the scope of this paper to address definitively why that is, but it would at least be reassuring to know that C20 activates the ISR in a wider range of cells, including ideally some primary, non-immortalized cells. In addition, the ISR is a complex, feedback-regulated response whose output varies depending on the time point examined. The in cellulo analysis in this paper is limited to reporter assays at 18 hours and qRT-PCR assays at 4 and 8 hours. A more extensive examination of the behavior of the relevant ISR mRNAs and proteins (eIF2, ATF4, CHOP, cell viability, etc.) for C20 across a more extensive time course would give the reader a clearer sense of how this molecule affects ISR output. I also find it a bit strange that the authors describe C20 as "demonstrat(ing) weak inhibition of ... PKR"-the measured IC₅₀ is ~4 μM, which is right around its EC₅₀ for GCN2 activation. This raises the confounding possibility that C20 would simultaneously activate GCN2 while inhibiting PKR. While perhaps inhibition of PKR is not relevant under the conditions when GCN2 would be activated either experimentally or therapeutically, examining in cells the effects of C20 on GCN2 and PKR across a dose range would shed light on whether this cross-reactivity is likely to be of concern.

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Author response:

We thank the editors and reviewers for their encouraging comments and constructive feedback. We will revise the text to enhance clarity as suggested. New experiments are planned to address questions raised regarding the time course of responses to the hit compounds. We also intend to examine additional endogenous readouts of the integrated stress response, including effects on translation. The effects of lead compound 20 will be examined in a wider range of cells, including primary cells.

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